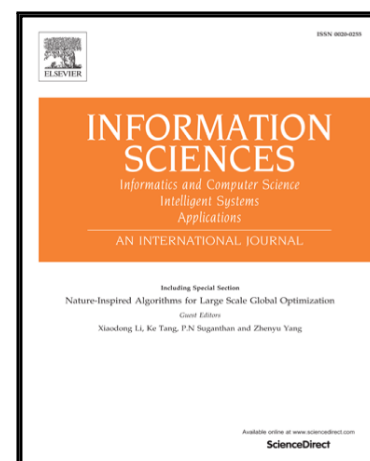


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Diagnostic Assessment of Deep Learning Algorithms for Diabetic Retinopathy Screening

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Abstract

Diabetic retinopathy (DR), the leading cause of blindness for working-age adults, is generally intervened by early screening to reduce vision loss. A series of automated deep-learning-based algorithms for DR screening have been proposed and achieved high sensitivity and specificity (>90%). However, these deep learning models do not perform well in clinical applications due to the limitations of the existing publicly available fundus image datasets. In order to evaluate these methods in clinical situations, we collected 13,673 fundus images from 9,598 patients. These images were divided into six classes by seven graders according to image quality and DR level. Moreover, 757 images with DR were selected to annotate four types of DR-related lesions. Finally, we evaluated state-of-the-art deep learning algorithms on collected images, including image classification, semantic segmentation and object detection. Although we obtain an accuracy of 0.8284 for DR classification, these algorithms perform poorly on lesion segmentation and detection, indicating that lesion segmentation and detection are quite challenging. In summary, we are providing a new dataset named DDR for assessing deep learning models and further exploring the clinical applications, particularly for lesion recognition.

Keywords: diabetic retinopathy, fundus image, deep learning, image classification, semantic segmentation

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1. Introduction

Diabetic retinopathy (DR), an eye disease caused by diabetes, is a leading cause of vision loss in working-age adults. It is reported that approximately 1/3 (34.6%) of people with diabetes have DR to some extent in the US, Europe and Asia [16]. It is also noted that 1 in 10 (10.2%) have vision-threatening DR [35]. It has been shown that vision loss resulting from DR can be prevented with timely treatment [32]. However, the early screening of DR remains a challenge for several reasons. First, diabetics are generally treated at endocrinology departments in hospitals, and fundus examination has long been ignored, resulting in delayed treatment for many patients. Second, the process of DR screening is time-consuming; consequently, only a certain number of patients can be processed each day. Third, the number of ophthalmologists cannot meet the growing demands around the world, particularly in developing regions [3].

Automated DR screening is an alternative solution to the above problems, with the advantages of high efficiency, low cost and minimal dependence on clinicians. Unlike conventional machine learning methods that rely heavily on feature engineering, deep learning algorithms automatically learn the most predictive representations in a manner of layer-by-layer feature combinations [20]. These methods have brought about a series of breakthroughs in the field of computer vision and have been applied to fundus image analysis, such as DR screening, lesion detection, and so forth [11, 2, 31, 10].

However, there are still huge gaps between research and practical applications for several reasons. 1) Automated DR screening algorithms only provide a DR grading result, which is not convincing for patients. We believe that this is the major factor that prevents deep learning methods from being applied in clinical applications since ophthalmologists generally evaluate the severity of DR based on the presence of lesions. 2) Publicly available DR datasets have several drawbacks. For example, the Messidor dataset only contains 1,200 fundus images [7], which is far from sufficient for deep neural networks. Kaggle DR [18] is the largest fundus image dataset, but it contains many ungradable images. Additionally, the performance on these datasets cannot reflect the real precision in clinical applications. 3) A large amount of data are required to

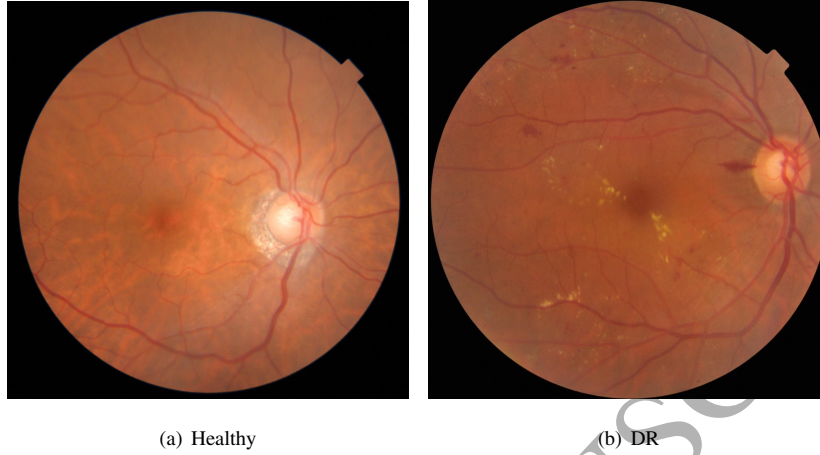


Figure 1: Example fundus images from our dataset.

30 train deep learning models for image classification, semantic segmentation and object detection. And the performance of deep learning models is highly related to the quality of fundus images.

Our work is motivated by solving the aforementioned problems. First, it is expected that an automated DR screening system should have the ability to identify abnormalities associated with DR, as is performed by human ophthalmologists. Thus, we evaluate popular deep-learning-based semantic segmentation models and detection models to recognize lesions. To achieve this goal, we annotated 757 fundus images with four types of lesions related to DR: microaneurysms (MA), soft exudates (SE), hard exudates (EX) and haemorrhages (HE). The experimental results show that both semantic segmentation models and detection models behave poorly. Specifically, the lesion segmentation model mAP scores are at most 0.3010, which is insufficient to meet clinical application requirements. Second, in order to evaluate the DR screening network from a closer clinical perspective, we collected 13,673 fundus images from 147 hospitals, covering 23 provinces in China. Our ophthalmologists classified the gradable images into 5 classes according to DR severity: none, mild, moderate, severe, and proliferative DR. And the sixth category is the images with poor quality. Figure 1 shows two fundus images from our dataset. We train and test popular classification networks on

these 13,673 fundus images, and the experimental results show that the average accuracy reaches 0.61 at most. Third, we provide a general-purpose DR dataset, called
 50 DDR, for DR screening, lesion segmentation and lesion detection. DDR consists of 13,673 fundus images with image-level annotations, and among them, 757 images are provided with pixel-level and bounding-box-level annotations. DDR is provided for researchers to implement and evaluate their models. Moreover, DDR provides both image-level and pixel-level annotations, which makes it possible to design multi-task
 55 deep learning models. For such models, both the DR level and lesions' locations can be given.

The main contributions of this paper include the following:

- A new fundus image dataset named DDR has been built and will be released for DR screening. To the best of our knowledge, DDR is the largest dataset
 60 that provides pixel-level and bounding-box annotations for microaneurysms, soft exudates, hard exudates and haemorrhages.
- DDR has been used to evaluate ten state-of-the-art deep learning models, including five classification models, two segmentation models and three detection models. Experimental results demonstrate that the performance of the model for
 65 small lesion recognition must be further improved to apply deep learning models to clinical practice.
- DDR provides multi-level annotations for a single image, i.e., image level, pixel level and bounding-box level. It can be used to design more robust and multi-task models, such as joint DR grading and lesion segmentation so that lesion
 70 segmentation can provide a diagnostic basis for DR grading.

2. Related Works

In this section, we review deep-learning-based DR screening algorithms and publicly available fundus image datasets.

2.1. Deep Learning in DR Screening

75 In recent years, deep learning has been applied to various tasks related to retinal fundus image analysis. Fu et al. proposed a novel Disc-aware Ensemble Network (DENet) for automatic glaucoma screening and reported an AUC of 0.9183 on the SCES dataset and an AUC of 0.8173 on the SINDI dataset [9]. Fu et al. also proposed a one-stage multi-label framework called M-Net to identify both the optic cup and optic disc. This framework achieved satisfactory glaucoma screening performances with calculated cup-to-disc ratio (CDR) values on both the ORIGA and SCES datasets [8].

80 In 2016, Gulshan et al. trained an Inception-v3 network to detect referable diabetic retinopathy (RDR). They reported a sensitivity of 97.5% and a specificity of 93.4% at the high-sensitivity operating point on the EyePACS-1 dataset [11]. Dsw et al. trained a deep learning system to detect RDR, vision-threatening DR, and related eye diseases. They further validated the system on 10 additional multiethnic datasets from different countries [31].

85 Abrmoff et al. presented a deep-learning-enhanced algorithm for the automated detection of DR. Their algorithm achieved significantly better performance than previously reported algorithms that did not employ deep learning [2]. Gargeya and Leng created a fully automated algorithm using deep learning for DR detection, and their algorithm demonstrated the potential application of deep learning in DR screening [10].

90 Dai et al. presented a clinical-report-guided multi-sieving convolutional neural network (MS-CNN) that integrates expert domain knowledge and image information to identify the potential microaneurysm regions [5]. Their method is semiautomatic rather than fully automatic, and it depends on expert knowledge in some sense.

95 Tan et al. proposed a 10-layer convolutional neural network to simultaneously segment and discriminate exudates, haemorrhages and microaneurysms [30]. How these models behave in real clinical situations remains to be explored since fundus images vary heavily due to different fundus cameras, photo settings and patients, among other factors. DDR is proposed for this assessment.

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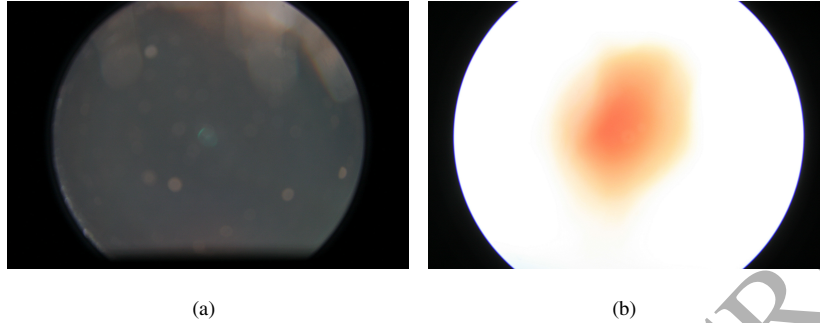


Figure 2: Sample images from the Kaggle DR dataset. These two images are both ungradable, but they are graded as no DR.

2.2. Related Datasets

2.2.1. Kaggle DR Dataset

The Kaggle website organized a DR detection challenge in 2015 [18]. The competition was sponsored by the California Healthcare Foundation. The Kaggle DR dataset consists of 88,702 colour fundus images, including 35,126 samples for training and 53,576 samples for testing. The images were captured under various conditions (e.g. resolutions) by various devices at multiple primary care sites throughout California and elsewhere. For each subject, two images of the left and right eyes were collected, with the same resolution. A clinician was asked to rate each image for the presence of DR with a scale of 0-4 according to the ETDRS scale [33]. Kaggle DR is the largest publicly available DR classification dataset, but it has several drawbacks. First, it does not include any manual annotations such as lesion contours or positions. Second, Figure 2 shows two ungradable images from Kaggle DR; these are indicative of the many poor-quality images in this dataset. However, analysing a poor-quality image makes no sense. Finally, each image in this dataset is graded by only one expert[18], which can potentially lead to annotation bias.

2.2.2. Messidor

The Messidor dataset [7] contains 540 normal images and 660 abnormal images. The Messidor project created this dataset to evaluate different lesion segmentation

120 methods for colour fundus images in the framework of DR screening and diagnosis. The images were captured using 8 bits per colour plane at 1440×960 , 2240×1488 or 2304×1536 pixels. Eight-hundred images were acquired with pupil dilation, and 400 images were acquired without dilation. These fundus images were acquired by 3 ophthalmologic departments using a colour video 3CCD camera on a Topcon TRC NW6
 125 non-mydratic retinograph with a 45-degree field of view (FOV). The DR grade and risk of macular edema have been provided by medical experts for each image. The DR severity was graded into four levels based on the presence and number of microaneurysms, haemorrhages, and neovascularization. Hard exudates were used to grade the risk of macular edema. There were no lesion annotations in the dataset. Although
 130 the Messidor dataset has been widely used to facilitate studies on computer-assisted diagnoses of DR, it is prone to overfitting when training with deep neural networks. Finally, its reference standard for DR grading is not consistent with the international standard, which limits its universal applicability.

2.2.3. IDRiD

135 The Indian Diabetic Retinopathy Image Dataset (IDRiD) is a fundus image dataset of an Indian population, and it consists of 516 images [23]. The images in this dataset were acquired using a Kowa VX-10 digital fundus camera with a 50-degree FOV. The images have a resolution of 4288×2848 pixels and are stored in jpg file format. First, medical experts graded the DR severity level on a scale from 0 (no apparent DR) to 4
 140 (severe DR) and the risk of diabetic macular edema (DME) on a scale from 0 (no DME) to 2 (severe DME) for the full set of 516 images. The image-level disease grading dataset was divided into a training set and a testing set, consisting of 413 (80%) and 103 (20%) images, respectively. Then, pixel-level annotations of typical DR-related lesions and optic disc were provided for 81 images with signs of DR. Along with the
 145 annotations presented above, the dataset provides centre pixel locations of the optic disc and fovea for all images. In summary, IDRiD is a high-quality dataset that provides both DR and DME pathologies and annotations of the major fundus structures, but the number of images in IDRiD is too small.

Other datasets are available for researchers in retinal image analysis. The DRIVE [28]

Table 1: Comparison between existing fundus image datasets and DDR.

Dataset	Total	Healthy	Images with DR	DR Grading Standard	Multiple Experts	Lesion Pixel-level Annotation	Number of Lesion Types	Lesion Detection Annotation
Kaggle DR	88702	65343	23359	international	No	No	-	No
Messidor	1200	540	660	local	Yes	No	-	No
IDRiD	516	168	348	international	Yes	Yes	4	No
e-ophtha_EX	82	35	47	-	No	Yes	1	No
e-ophtha_MA	381	233	148	-	No	Yes	1	No
DRiDB	50	14	36	international	Yes	No	-	No
DIARETDB1	89	5	84	-	Yes	No	-	No
DDR	13673	6266	6256	international	Yes	Yes	4	Yes

and STARE [13] datasets focus on blood vessel segmentation. The Retinopathy Online Challenge (ROC) microaneurysm dataset [22] was published for a microaneurysm detection competition arranged by the University of Iowa in 2009, and the dataset consists of 50 training images and 50 test images. The e-ophtha dataset [6] is concerned with exudates and microaneurysms. This dataset consists of two subsets, named e-ophtha MA and e-ophtha EX. The e-ophtha EX subset contains 47 images with 12,278 exudates, as well as 35 healthy images. The e-ophtha MA subset contains 148 images with 1,306 microaneurysms, as well as 233 healthy images. The DIARETDB1 dataset [19] contains 89 images, of which 5 images represent healthy retinas, while the other 84 have some signs of DR. The images were annotated by four experts for the presence of MA, HE, EX and SE. DRiDB [24] is a comprehensive dataset in which DR severity is graded, and all fundus structures and pathologies are annotated. However, lesion annotations are not performed at the pixel level in either the DIARETDB1 or the DRiDB datasets. Table 1 compares several publicly available fundus image datasets. Table 2 shows a comparison of several datasets that have pixel-level DR-related lesion annotations.

Compared to currently available datasets, DDR contains more data, which helps to reduce overfitting. The appearance of poor-quality images is inevitable in clinical practice, and it is meaningless for such images to be graded. To improve the DR detection accuracy, in addition to five levels of DR severity, we classify the ungradable images

Table 2: Comparison between DR lesion segmentation annotation datasets.

Dataset	Total	HE	MA	SE	EX
IDRiD	81	80	81	40	81
e-optha_EX	47	-	-	-	47
e-optha_MA	148	-	148	-	-
DDR	757	601	570	239	486

into one class. DDR is annotated by multiple experts, which ensures the reliability of the dataset. The grading standard is consistent with the international standards published by the International Council of Ophthalmology (ICO), which makes DDR generally applicable. To achieve a quantitative analysis of lesions, four types of DR-related lesions are annotated on 757 DR images at the pixel level. Occasionally, researchers are only concerned with lesion positions; thus, we also provide bounding-box annotations for lesion detection. Table 1 shows that DDR is the second largest dataset for DR grading. Additionally, no other dataset contains DR lesion bounding-box annotations. Table 2 shows that only a few datasets provide pixel-level annotations of lesions, and DDR is the largest of them. More importantly, we provide all four types of lesions associated with DR, whereas the e-optha dataset only annotates microaneurysms and hard exudates.

3. Materials

3.1. Dataset Collection

To ensure the general applicability of the dataset, data collection should be conducted according to the principle of uniform distributions of patients' age, gender and living area and a comprehensive coverage of collection equipment.

We collected 13,673 colour fundus images from 147 hospitals from 2016 to 2018, covering 23 provinces in China, 84 of which are grade-A tertiary hospitals. All images have been desensitized for common use. These images are provided by 9,598 patients whose ages ranged from 1 to 100, with an average age of 54.13. Among these images, 48.23% are from male patients, and 51.77% are from female patients.

These fundus images are captured using 42 types of fundus cameras with a 45 degree FOV, mainly Topcon D7000, Topcon TRC NW48, Nikon D5200 and Canon CR 2 cameras. Data are collected using single-view images and photographed according to the following three principles. First, the centre of a single fundus image should be located at the middle point of the line connecting the optic disc and the fovea. The central fovea or optic disc centre should be horizontal in the single fundus image. Second, fundus photography requires accurate focus. In healthy humans, the upper disc surface, retinal main blood vessels, retinal nerve fibre layer, macular fovea and other structures should be clearly identifiable. Additionally, lesions should be clear and legible for patients with DR. Third, the fundus image should be moderately exposed. The interface between the optic disc and the optic cup, the small blood vessels on the surface of the optic disc, and the normal retinal nerve fiber layer are clearly distinguishable. Finally, when taking a fundus photograph, the patient should have their eye opened wide and should not blink. Hair and eyelashes are not allowed in the field at the moment of capturing the image.

3.2. Annotation Method

The dataset provides three types of annotations: image-level DR grading annotations, pixel-level annotations, and bounding-box annotations of lesions associated with DR.

DR Grading Annotation. All images are graded to determine the severity of DR by ophthalmologists according to the International Classification of Diabetic Retinopathy [16]. However, in clinical practice, poor-quality images may inevitably appear due to exposure or other reasons. Image quality has a significant impact on the performance of the DR screening algorithms. Thus, we made a distinction in image quality and classified ungradable images into a single category. Images with a blurring degree of more than 70% and without clearly visible lesions are considered ungradable. In other words, these images in DDR are divided into six classes: no DR, mild nonproliferative DR, moderate nonproliferative DR, severe nonproliferative DR, proliferative DR, and ungradable.

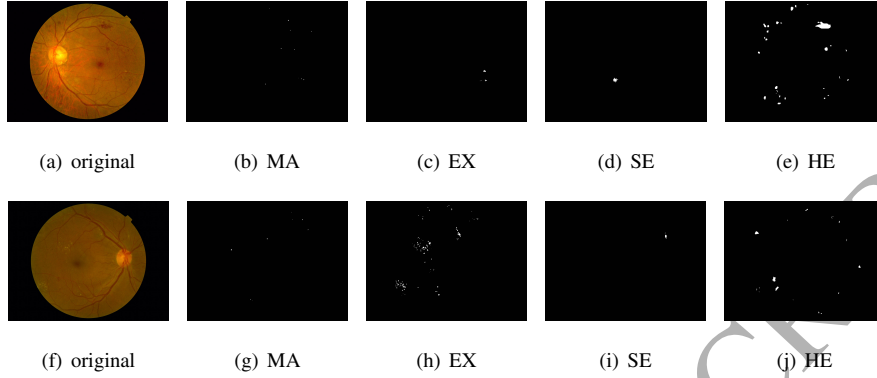


Figure 3: Sample images with pixel-level annotations for lesions. Original images and ground truth for MA, EX, SE and HE are shown from left to right.

Professional graders have been trained by ophthalmologists of Beijing Tongren Hospital and Beijing Institute of Ophthalmology. These graders are able to identify all types of common eye diseases. Moreover, they attended the Chinese version of the international GREG (Gloucestershire Retinal Education Group) course of Grading in Diabetic Retinopathy Screening. After completing the course, they passed the examination and earned the Certificate of Grading in Diabetic Retinopathy Screening. Seven trained graders are responsible for grading the level of DR. Each image was graded by approximately four graders, and each grader is assigned to process approximately 8,000 images. The seven graders vote to evaluate images in question, and the final classification was determined by majority voting. If the seven graders could not determine a classification result, they will consult more experienced specialists (more than 30 years of experience). All fundus images are checked and confirmed before being stored. Information about the ophthalmologist is recorded for each image, which can track the grading ophthalmologist who is initially assigned.

Pixel-Level Annotations. We select 757 fundus images to annotate DR lesions, and each image is annotated in two ways, including pixel-level annotations and bounding-box annotations. These selected images are all gradable, and the severity of DR varies from 1 (mild DR) to 4 (proliferative DR).

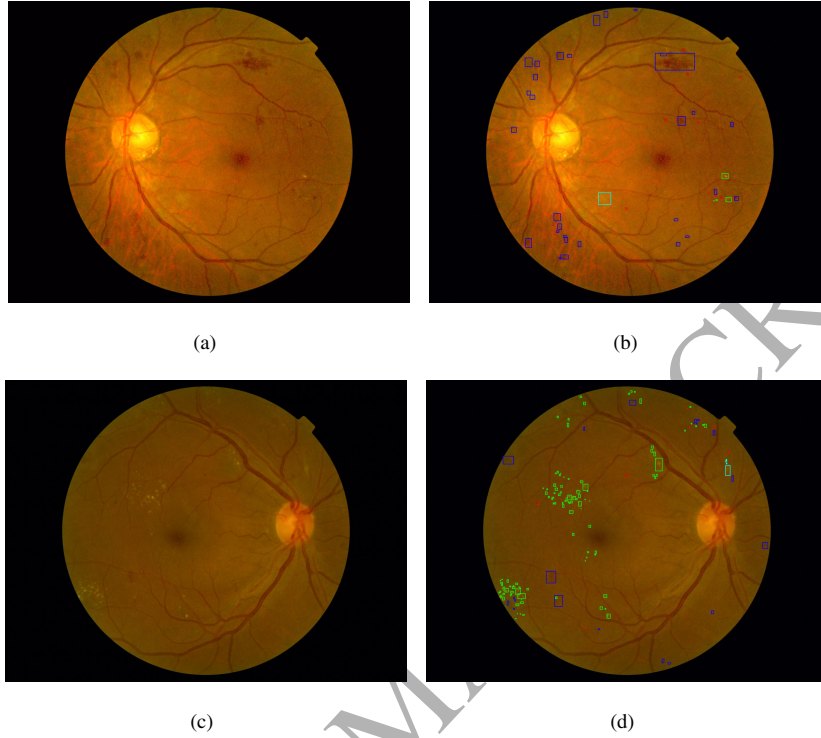


Figure 4: Sample images with bounding-box annotations for lesions. The original images are on the left. The annotated images are on the right. Green marks represent EX, red marks represent MA, blue marks represent HE, and light blue marks represent SE.

Six annotators perform pixel-level annotations for four lesions, including MA, EX, SE, and HE, using special software. The annotation tool has a function to automatically identify lesions. The annotators focus on whether lesions and boundaries are correctly identified. The annotators manually annotate lesions that are not identified by the annotation tool and remove false lesions. After an image is annotated, it is saved as a MAT file and a colour image; four lesions can be distinguished by 4 colours. Finally, the ground truth is extracted from the MAT files. Figure 3 shows samples of the original images and ground truth with pixel-level annotations for lesions.

Bounding-box Generation. The bounding-box annotations are automatically generated from the pixel-level annotations. First, we found all connected components for

Table 3: The number of samples of each class in the training, validation and testing sets.

	Training	Validation	Testing	Total
No DR	3133	1253	1880	6266
Mild	315	126	189	630
Moderate	2238	895	1344	4477
Severe	118	47	71	236
Proliferative	456	182	275	913
Ungradable	575	230	346	1151
Total	6835	2733	4105	13673

each segmentation annotation. Second, for each connected component, the smallest
 250 rectangles are generated, containing the area that is taken as the bounding-box annotation of the lesion. Figure 4 shows sample images with bounding-box annotations for lesions.

3.3. Dataset Splits

To ensure that the training data and test data distributions are approximately matched,
 255 we randomly selected half of the original images as a training set, 2/10 as a validation set, and 3/10 as a testing set. We will publicly provide all original images with the label or ground truth. Table 3 shows the distribution of training, validation and test images per class for DR grading.

4. Experiments

260 DDR involves three types of tasks in computer vision: image classification, semantic segmentation, and object detection.

4.1. DR Grading

Preprocessing. First, we cropped out all backgrounds and resized the images to 512×512 pixels. Note that the classes are heavily imbalanced in DDR. For example, no DR samples account for 45.84%, whereas mild DR samples only account for 4.61%. Inspired
 265 by the solution of team o_O, who ranked second in the Kaggle DR detection competition, we applied a resampling strategy to all classes in the training set such that all

Table 4: DR grading results for various deep neural networks on the validation set.

Model	No DR	Mild	Moderate	Severe	Proliferative	Ungradable	OA	AA	kappa
VGG-16	0.9689	0.0873	0.7173	0.5532	0.7692	0.8826	0.8182	0.6631	0.7280
ResNet-18	0.9505	0.0714	0.7564	0.3404	0.7747	0.8304	0.8141	0.6206	0.7220
GoogLeNet	0.9585	0.0714	0.7542	0.4468	0.7473	0.8261	0.8185	0.6340	0.7286
DenseNet-121	0.8252	0.3016	0.6637	0.5320	0.7527	0.8826	0.7431	0.6596	0.6158
SE-BN-Inception	0.9401	0.0714	0.8190	0.3617	0.6978	0.8696	0.8284	0.6266	0.7438

Table 5: DR grading results for various deep neural networks on the test set.

Model	No DR	Mild	Moderate	Severe	Proliferative	Ungradable	OA	AA	kappa
VGG-16	0.9537	0.0423	0.5625	0.3944	0.6436	0.9422	0.7520	0.5898	0.6297
ResNet-18	0.9548	0.0582	0.6220	0.3662	0.5818	0.9133	0.7659	0.5827	0.6500
GoogLeNet	0.9574	0.0265	0.5759	0.3380	0.5782	0.9162	0.7498	0.5654	0.6264
DenseNet-121	0.8930	0.2275	0.5751	0.4085	0.6364	0.9306	0.7357	0.6119	0.6053
SE-BN-Inception	0.9452	0.0476	0.6458	0.1268	0.5818	0.9046	0.7639	0.5418	0.6472

classes are represented equally on average. Then, data augmentation was adopted to reduce overfitting and increase the performance of the algorithm. In our experiments, we applied translation, stretching, rotation, flipping, and colour augmentation to the labelled dataset.

Results and Analysis.

For DR grading, VGG-16 [27], ResNet-18 [12], GoogLeNet [29], DenseNet-121 [15], and SE-BN-Inception [14] are selected as benchmarking models. All these networks are fine-tuned from ImageNet pretrained weights. All training and testing images fed into the networks are resized to 224×224 pixels. Our network is implemented in the Caffe [17] framework on a Linux machine equipped with an Nvidia Tesla K40C GPU.

Table 4 summarizes the accuracy for each class, the overall accuracy (OA), the average accuracy (AA), and the kappa scores of the models on the validation set. Table 5 presents the results on the test set. The experimental results demonstrate that the SE-BN-Inception network shows the highest overall accuracy and kappa score on the validation set but a comparatively low average accuracy. All these networks exhibit poor performance on samples with mild and severe DR, suggesting that these models

Table 6: AP and IoU metrics of lesion segmentation on the validation set.

Models	AP					IoU				
	MA	EX	SE	HE	mAP	MA	EX	SE	HE	mIoU
HED	0.0600	0.2709	0.1612	0.1885	0.1702	0.0204	0.0948	0.0379	0.2183	0.0929
DeepLab-v3+	0.0390	0.4375	0.4451	0.4053	0.3317	0.0429	0.2910	0.2756	0.2819	0.2229

Table 7: AP and IoU metrics of lesion segmentation on the test set.

Models	AP					IoU				
	MA	EX	SE	HE	mAP	MA	EX	SE	HE	mIoU
HED	0.0311	0.3221	0.2271	0.0544	0.1587	0.0110	0.1874	0.0782	0.0524	0.0883
DeepLab-v3+	0.0205	0.5634	0.4359	0.1842	0.3010	0.0325	0.3118	0.2295	0.1425	0.1791

are weak in learning these two types of samples. Meanwhile, the number of these two types of samples is relatively small. Furthermore, microaneurysms in mild DR are difficult to identify, while samples of severe DR are easily misclassified as moderate DR. Note that although DenseNet-121 has the lowest overall accuracy on both the validation and test sets, it achieves a relatively high accuracy on samples with mild DR and severe DR. These results suggest that DenseNet-121 performs better than the other models on classes that only contain a small number of samples.

4.2. DR Lesion Segmentation

To evaluate the difficulty of DR lesion segmentation, we tested two semantic segmentation models on the DDR dataset: DeepLab-v3+ [4] and HED [34]. We used a TensorFlow [1] implementation of DeepLab-v3+, initialized from Pascal VOC. And HED is implemented in the Caffe framework.

Tables 6 and 7 present the average precision (AP) and intersection over union (IoU) metrics on each type of lesion and the average value on the validation set and test set. The results suggest that MAs are too small for these two models to identify. As the marker of mild DR, MAs generally cover only a few pixels and are easily missed or misclassified. In general, DeepLab-v3+ has shown far better performance than HED. Figures 5 and 6 show the segmentation results obtained using DeepLab-v3+ and HED, respectively. The precision-recall curves on the validation set and test set are displayed

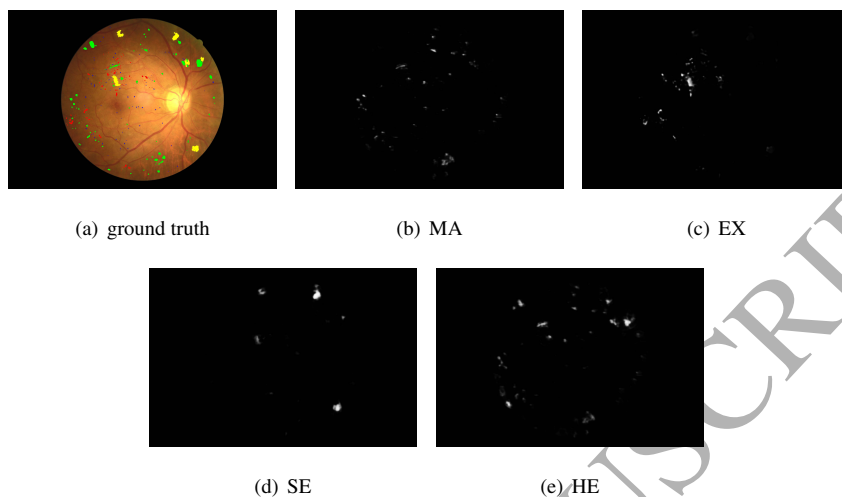


Figure 5: Segmentation results using DeepLab-v3+. Ground truth (red markings denote EX, green markings denote HE, blue markings denote MA, and yellow markings denote SE) and segmentation results for MA, EX, SE and HE are shown from left to right.

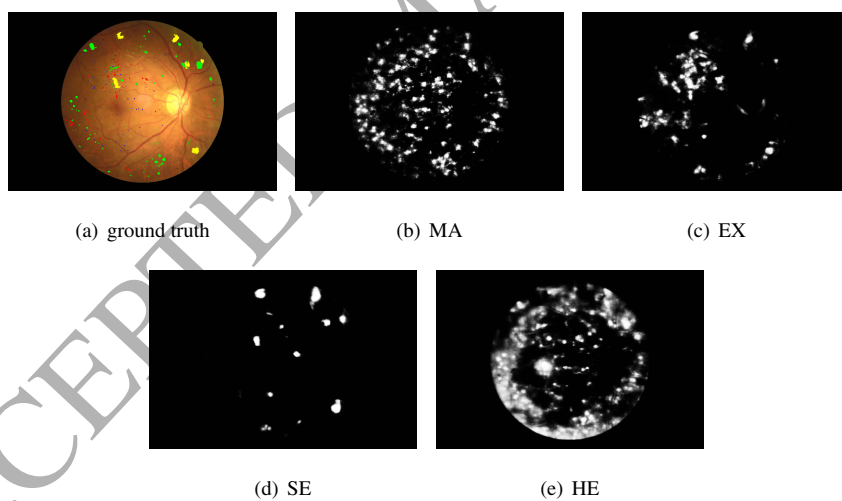
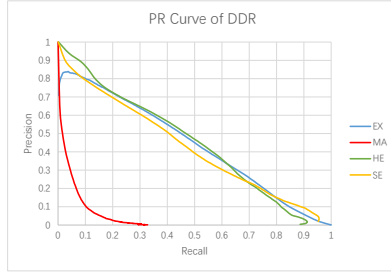
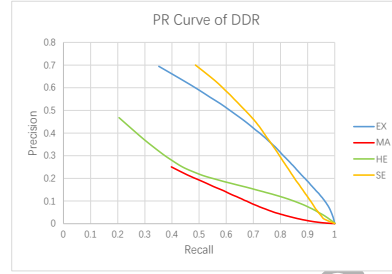


Figure 6: Segmentation results using HED. Ground truth (red markings denote EX, green markings denote HE, blue markings denote MA, and yellow markings denote SE) and segmentation results for MA, EX, SE and HE are shown from left to right.

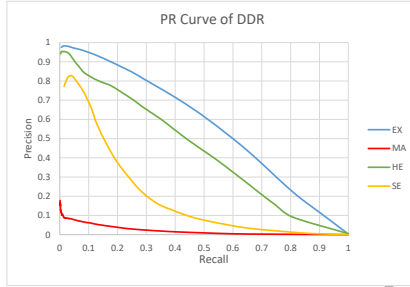


(a) DeepLab-v3+

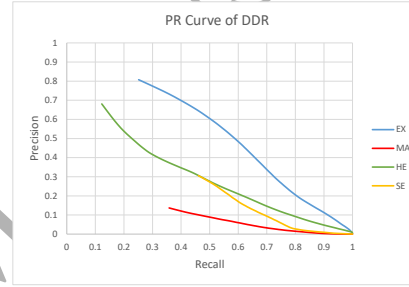


(b) HED

Figure 7: Precision-recall curves for each type of lesion using DeepLab-v3+ and HED on the validation set.



(a) DeepLab-v3+



(b) HED

Figure 8: Precision-recall curves for each type of lesion using DeepLab-v3+ and HED on the test set.

in Figures 7 and 8, respectively. Figure 9 shows the segmentation results in some regions.

Overall, DeepLab-v3+ performs better than HED, but it is far from meeting clinical requirements. The experimental results show that there is still considerable room for improvement in lesion segmentation in fundus images, particularly for MAs.

4.3. DR Lesion Detection

We tested the state-of-the-art object detection models to evaluate their performance on the DDR dataset, including Faster RCNN [26], SSD [21] and YOLO [25]. We implemented these models using the Caffe framework and fine-tuned the weights pre-

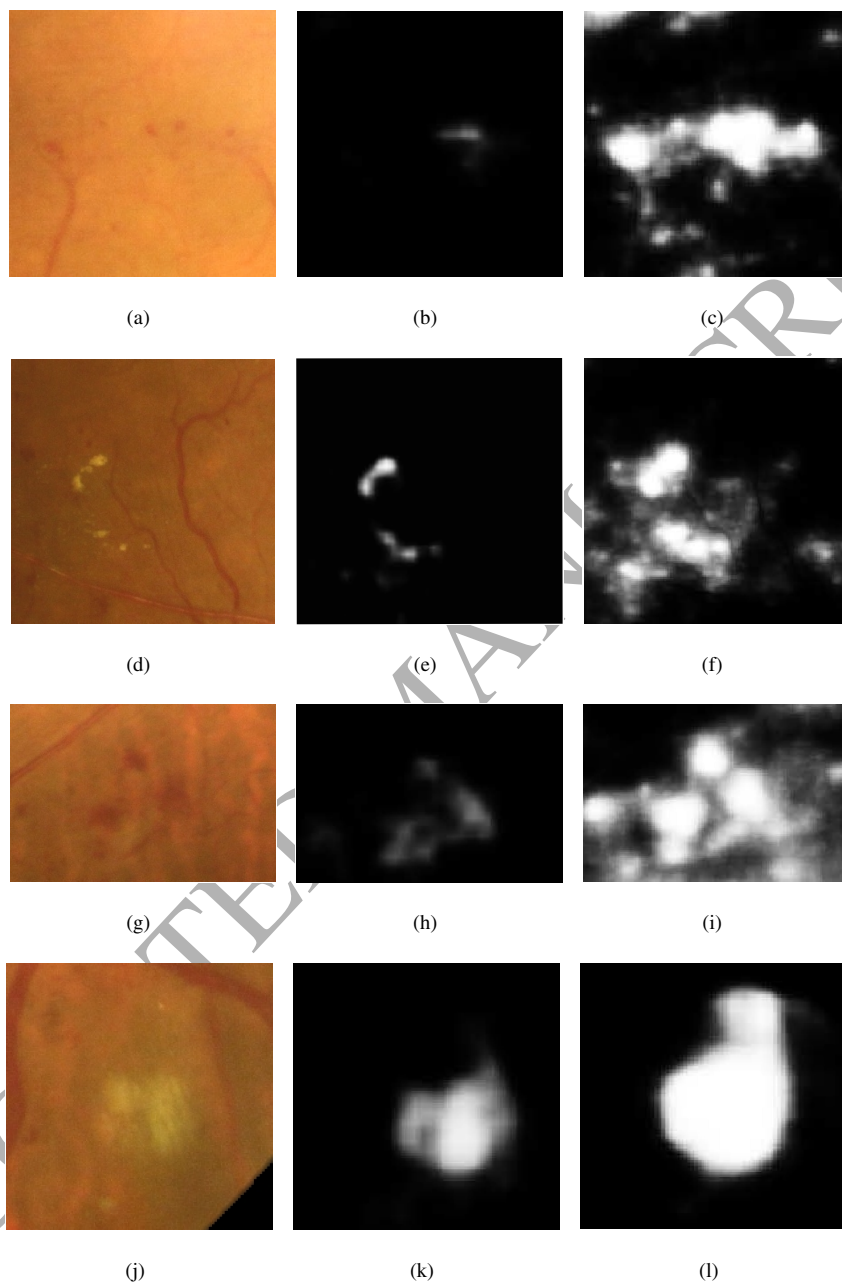


Figure 9: Segmentation results in some regions. The first column represents the original image. The second column represents the segmentation results of DeepLab-v3+. The third column represents the segmentation results of HED. MA, EX, HE and SE are shown from top to bottom.

Table 8: AP and IoU metrics of lesion detection on the validation set.

Models	AP					IoU				
	MA	EX	SE	HE	mAP	MA	EX	SE	HE	mIoU
SSD	0.000082	0.000091	0.022727	0.000777	0.005919	0.001905	0.001323	0.002181	0.000589	0.001500
YOLO	0	0.003953	0	0.010101	0.003513	0.000013	0.000409	0	0.001705	0.000532
Faster RCNN	0.009091	0.030303	0.243371	0.087059	0.092456	0.008021	0.023071	0.166047	0.077207	0.068586

Table 9: AP and IoU metrics of lesion detection on the test set.

Models	AP					IoU				
	MA	EX	SE	HE	mAP	MA	EX	SE	HE	mIoU
SSD	0.000155	0.000224	0	0.005682	0.001515	0.001524	0.002874	0	0.000375	0.001193
YOLO	0	0.001271	0	0.010909	0.003045	0.000030	0.000172	0	0.000841	0.000261
Faster RCNN	0.000411	0.001404	0	0.001783	0.000900	0.001420	0.023302	0	0.003230	0.006988

trained on the Pascal VOC dataset. The AP and IoU metrics on the validation set and test set are shown in Tables 8 and 9, and the AP metric is calculated with 0.1 IoU.

315 These models achieve only poor performances, suggesting that lesion detection in fundus images is extremely difficult and poses a challenge to researchers.

5. Discussion

Although deep learning has shown great potential in medical image analysis, it has several limitations. First, existing state-of-the-art algorithms behave well in DR classification but rather poorly in lesion segmentation and detection. As we all know, a DR prediction result alone is not convincing; therefore, there is a large gap between research and clinical application. Second, we have serious concerns about the generalization of deep learning methods. Although such methods could achieve impressive performance in one specific dataset, it is unknown how they may perform on other datasets. Finally, as a branch of machine learning, deep learning automatically extracts the most discriminative features from training samples. However, what features are extracted to make predictions remains unknown. Therefore, there is a lack of sufficient confidence to provide a diagnostic report. In summary, more explorations about deep learning methods for DR screening are needed.

330 6. Conclusions and future work

In this paper, we assessed the state-of-the-art deep learning models for DR classification, DR lesion segmentation, and DR lesion detection. To conduct this assessment, we compiled a new fundus image dataset that is applicable for DR screening. To the best of our knowledge, DDR is the first representative fundus image dataset for a Chinese population. Furthermore, DDR is the only one that considers lesion detection, and it is the largest one for lesion segmentation. Although DDR is the second largest for DR grading, we provide more accurate annotations. The experimental results show that lesion segmentation and detection are much more difficult than DR classification. The DDR dataset is available at <https://github.com/nkicsl/DDR-dataset>.

340 In the future, we plan to further enrich and expand our dataset, including the populations covered, disease types, etc. Nonetheless, lesion segmentation and detection in fundus images remain a challenge and better algorithms need to be developed. We plan to design a framework that combines DR grading and lesion segmentation. Accurate lesion segmentation can improve the classification accuracy because DR is graded according to lesion appearance.

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